Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Chapter 1
Decisions Under Risk: Probability and Expectation

There are as many different reasons for playing poker as there are players who play the game. Some play for social reasons, to feel part of a group or "one of the guys," some play for recreation, just to enjoy themselves. Many play for the enjoyment of competition. Still others - lay to satisfy gambling addictions or to cover up other pain in their lives. One of the difficulties of taking a mathematical approach to these reasons is that it's difficult to quantify the value of having fun or of a sense of belonging.

In addition to some of the more nebulous and difficult to quantify reasons for playing poker, there may also be additional financial incentives not captured within the game itself. For example, the winner of the championship event of the World Series of Poker is virtually guaranteed to reap a windfall from endorsements, appearances, and so on, over and above the large first prize.

There are other considerations for players at the poker table as well; perhaps losing an additional hand would be a significant psychological blow. While we may criticize this view as irrational, it must still factor into any exhaustive examination of the incentives to play poker. Even if we restrict our inquiry to monetary rewards, we find that preference for money is non-linear. For most people, winning five million dollars is worth much more (or has much more ***utility)*** than a 50°/o chance of winning ten million; five million dollars is life-changing money for most, and the ***marginal value*** of the additional five million is much smaller.

In a broader sense, all of these issues are included in the ***utility theory*** branch of economics. Utility theorists seek to quantify the preferences of individuals and create a framework under which financial and non-financial incentives can be directly compared. In reality, it is utility that we seek to maximize when playing poker (or in fact, when doing any tiring). However, the use of utility theory as a basis for analysis presents a difficulty; each individual has his own utility curves and so general analysis becomes extremely difficult.

In this book, we will therefore refrain from considering utility and instead use money won inside the game as a proxy for utility. In the bankroll theory section in Part IV, we will take an in-depth look at certain meta-game considerations, introduce such concepts as ***risk of ruin, the Kelly criterion,*** and ***certainty equivalent.*** All of these are measures of risk that have primarily to do with factors outside the game. Except when expressly stated, however, we will take as a premise that players are adequately bankrolled for the games they are playing in, and that their sole purpose is to maximize the money they will win by making the best decisions at every point.

Maximizing total money won in poker requires that a player maximize the ***expected value*** of his decisions. However, before we can reasonably introduce this cornerstone concept, we must first spend some time discussing the concepts of probability that underlie it. The following material owes a great debt to Richard Epstein's text *The Theory of Gambling and Statistical Logic* (1967), a valuable primer on probability and gambling.

Probability

Most of the decisions in poker take place under conditions where the outcome has not yet been determined. When the dealer deals out the hand at the outset, the players' cards are unknown, at least until they are observed. Yet we still have some information about the contents of the other players' hands. The game's rules constrain the contents of their hands-while a player may hold the jack-ten of hearts, he cannot hold the ace-prince of billiard tables, for example. The composition of the deck of cards is set before starting and gives us information about the hands.

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Consider a holdem hand. What is the chance that the hand contains two aces? You may know the answer already, but consider what the answer means. What if we dealt a million hands just like this? How many pairs of aces would there be? What if we dealt ten million? Over time and many trials, the ratio of pairs of aces to total hands dealt will converge on a particular number. We define ***probability*** as this number. Probability is the key to decision-making in poker as it provides a mathematical framework by which we can evaluate the likelihood of uncertain events.

If n trials of an experiment (such as dealing out a holdem hand) produce *no* occurrences of an event *x,* we define the probability *p* of *x* occurring *p{x)* as follows:

$$p(x) = \lim_{n\to\infty} \frac{n_0}{n} \qquad \textbf{(1.1)}$$

Now it happens to be the case that the likelihood of a single holdem hand being a pair of aces is $^1/_{221}$. We could, of course, deterrnine this by dealing out ten billion hands and observing the ratio of pairs of aces, to total hands dealt. Tins, however, would be a lengthy and difficult process, and we can do better by breaking the problem up into components. First we consider just one card. What is the probability that a single card is an ace? Even this problem can be broken down further - what is the probability that a single card is the ace of spades?

This final question can be answered rather directly. We make the following assumptions:

- There are fifty-two cards in a standard deck.
- Each possible card is equally likely.

Then the probability of any particular card being the one chosen is $^1/_{52}$. If the chance of the card being the ace of spades is $^1/_{52}$, what is the chance of the card being any ace? This is equivalent to the chance that the card is the ace of spades OR that it is the ace of hearts OR that it is the ace of diamonds OR that it is the ace of clubs. There are four aces in the deck, each with a $^1/_{52}$ chance of being the card, and summing these probabilities, we have:

$$p(A) = (4)\left(\frac{1}{52}\right)$$

$$p(A) = \left(\frac{1}{13}\right)$$

We can sum these probabilities directly because they are ***mutually exclusive;*** that is, no card can simultaneously be both the ace of spades and the ace of hearts. Note that the probability $^1/_{13}$ is exactly equal to the ratio (number of aces in the deck)/(number of cards total). This relationship holds just as well as the summing of the individual probabilities.

Independent Events

Some events, however, are not mutually exclusive. Consider for example, these two events:

1. The card is a heart
2. The card is an ace.

If we try to figure out the probability that a single card is a heart OR that it is an ace, we find there are thirteen hearts in the deck out of fifty-cards, so the chance that the card is a heart is $^1/_4$. The chance that the card is an ace is, as before, $^1/_{13}$. However, we cannot simply add these probabilities as before, because it is possible for a card to both be an ace and a heart.
There are two types of relationships between events. The first type are events that have no effect

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

on each other. For example, the closing value of the NASDAQ stock index and the value of the dice on a particular roll at the craps table in a casino in Monaco that evening are basically unrelated events; neither one should impact the other in any way that is not negligible. If the probability of both events occurring equals the product of the individual probabilities, then the events are said to be ***independent.*** The probability that both A and B occur is called the ***joint probability*** of A and B.

In this case, the joint probability of a card being both a heart and an ace is $(^1/_{13})$ $(^1/_4)$, or $^1/_{52}$. This is because the fact that the card is a heart does not affect the chance that it is an ace - all four suits have the same set of cards.

Independent events are not mutually exclusive except when one of the events has probability zero. In this example, the total number of hearts in the deck is thirteen, and the total of aces in the deck is four. However, by adding these together, we are double-counting one single card (the ace of hearts). There are actually thirteen hearts and three other aces, or if you prefer, four aces, and twelve other hearts. It turns out that the general application of this concept is that the probability that at least one of two mutually non-exclusive events A and B will occur is the sum of the probabilities of A and B minus the joint probability of A and B. So the probability of the card being a heart ***or*** an ace is equal to the chance of it being a heart $(^1/_4)$ plus the chance of it being an ace minus the chance of it being both $(^1/_{52})$, or $^4/_{13}$. This is true for all events, independent or dependent.

Dependent Events

Some events, by contrast, do have impacts on each other. For example, before a baseball game, a certain talented pitcher might have a 3% chance of pitching nine innings and allowing no runs, while his team might have a 60% chance of winning the game. However, the chance of the pitcher's team winning the game and him also pitching a shutout is obviously not 60% times 3%. Instead, it is very close to 3% itself, because the pitcher's team will virtually always win the game when he accomplishes this. These events are called ***dependent.*** We can also consider the ***conditional probability*** of A given B, which is the chance that ***if*** B happens, A will also happen. The probability of A and B both occurring for dependent events is equal to the probability of A multiplied by the conditional probability of B given A. Events are independent if the conditional probability of A given B is equal to the probability of A alone.

Summarizing these topics more formally, if we use the following notation:

$p(A \cup B)$ = Probability of *A* or *8* occurring.
$p(A \cap B)$ = Probability of *A* and *B* occurring.
$p(A|B)$ = Conditional probability of ***A*** occurring given ***B*** has already occurred.

The ∪ and ∩ notations are from set theory and formally represent "union" and "intersection." We prefer the more mundane terms "or" and "and." Likewise, | is the symbol for "given," so we pronounce these expressions as follows:

$p(A \cup B)$ = “p of A or B”
$p(A \cap B)$ = “p of A and B”
$p(A|B)$ = “p of A given B”

Then for mutually exclusive events:

$$p(A \cup B) = p(A) + p(B) \qquad \textbf{(1.2)}$$

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

For independent events:
$p(A \cap B) = p(A)p(B)$ **(1.3)**

For all events:
$p(A \cup B) = p(A) + p(B) - p(A \cap B)$ **(1.4)**

For dependent events:
$p(A \cap B) = p(A)p(B|A)$ **(1.5)**

Equation 1.2 is simply a special case of Equation 1.4 for mutually exclusive events, $p(A \cap B) = 0$. Likewise, Equation 1.3 is a special case of Equation 1.5, as for independent events, $p(B|A) = p(B)$**.** Additionally, if $p(B|A) = p(B)$**.,** then $p(A|B) = p(A)$**.**

We can now return to the question at hand. How frequently will a single holdem hand dealt from a full deck contain two aces? There are two events here:

- A: The first card is an ace.
- B: The second card is an ace.

p(A) = $^1/_{13}$, and *p(B)* = $^1/_{13}$ as well. However, these two events are dependent, if *A* occurs (the first card is an ace), then it is less likely that *B* will occur, as the cards are dealt without replacement. So $p(B|A)$ is the chance that the second card is an ace given that the first card is an ace. There are three aces remaining, and fifty-one possible cards, so $p(B|A) = {}^3/_{51}$, or $^1/_{17}$.

$p(A \cap B) = p(A)p(B|A)$
$p(A \cap B) = ({}^1/_{13})({}^1/_{17})$
$p(A \cap B) = {}^1/_{221}$

There are a number of other simple properties that we can mention about probabilities. First, the probability of any event is at least zero and no greater than one. Referring back to the definition of probability, n trials will never result in more than n occurrences of the event, and never less than zero occurrences. The probability of an event that is certain to occur is one. The probability of an event that never occurs is zero. Tire probability of an event's ***complement*** -that is, the chance that an event does not occur, is simply one minus the event's probability.

Summarizing, if we use the following notation:

$p(\overline{A}) =$ Probability that *A* does not occur.

C = a certain event
I = an impossible event

Then we have:

$0 \leq p(A) \leq 1$ for any A **(1.6)**
p(C) = 1 **(1.7)**
p(I) = 0 **(1.8)**
$p(A) + p(\overline{A}) = 1$ **(1.9)**

Equation 1.9 can also be restarted as:
$p(A) = 1 - p(\overline{A})$ **(1.10)**

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

We can solve many probability problems using these rules
Some common questions of probability are simple, such as the chance of rolling double sixes on two dice, hi terms of probability, this can be stated using equation 1.3, since the die rolls are independent. Let *p(A)* be the probability of rolling a six on the first die and *p(B)* be the probability of rolling a six on the second die. Then:

$p(A \cap B) = p(A)p(B)$
$p(A \cap B) = (1/6)(1/6)$
$p(A \cap B) = 1/36$

Likewise, using equation 1.2, the chance of a single player holding aces, kings, or queens becomes:

$p(\text{AA}) = 1/221$
$p(\text{KK}) = 1/221$
$p(\text{QQ}) = 1/221$
$p(\{\text{AA, KK, QQ}\}) = p(\text{AA}) + p(\text{KK}) + p(\text{QQ}) = 3/221$

Additionally we can solve more complex questions, such as:

How likely is it that a suited hand will flop a flush?

We hold two of the flush suit, leaving eleven in the deck. All three of the cards must be of the flush suit, meaning that we have *A* — the first card being a flush card, *B* — the second card being a flush card given that the first card is a flush card, and C= the third card being a flush card given than both of the first two are flush cards.

$p(A) = 11/50$ (two cards removed from the deck in the player's hand)

$p(B|A) = 10/49$ (one flush card and three total cards removed)

$p(C|(A \cap B)) = 9/48$ (two flush cards and four total cards removed)

Applying equation 1.5, we get:

$p(A \cap B) = p(A)p(B|A)$
$p(A \cap B) = (1/50)(10/49)$
$p(A \cap B) = 11/245$

Letting D = $(A \cap B)$, we can use equation 1.5 again:

$p(D \cap C) = p(D)p(C|D)$
$p(A \cap B \cap C) = p(A \cap B)p(C|(A \cap B))$
$p(A \cap B \cap C) = (11/245)(9/48)$
$p(A \cap B \cap C) = 33/3920$, or a little less than 1%.

We can apply these rules to -virtually any situation, and throughout the text we will use these properties and rules to calculate probabilities for single events.

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Probability Distributions

Though single event probabilities are important, it is often the case that they are inadequate to fully analyze a situation. Instead, it is frequently important to consider many different probabilities at the same time. We can characterize the possible outcomes and their probabilities from an event as a probability distribution.

Consider a fair coin flip. The coin flip has just two possible outcomes - each outcome is mutually exclusive and has a probability of $^1/_2$. We can create a probability distribution for the coin flip by taking each outcome and pairing it with its probability. So we have two pairs: (heads, $^1/_2$) and (tails, $^1/_2$).

If *C* is the probability distribution of the result of a coin flip, then we can write this as:

C = {(heads, $^1/_2$), (tails, $^1/_2$)}

Likewise, the probability distribution of the result of a fair six-sided die roll is:
D = {(1,$^1/_6$), (2,$^1/_6$), (3, $^1/_6$), (4, $^1/_6$), (5, $^1/_6$), (6, $^1/_6$)}

We can construct a discrete probability distribution for any event by enumerating an exhaustive and mutually exclusive list of possible outcomes and pairing these outcomes with their corresponding probabilities.

We can therefore create different probability distributions from the same physical event. From our die roll we could also create a second probability distribution, this one the distribution of the odd-or-evenness of the roll:

D' = {(odd, $^1/_2$), (even, $^1/_2$)}

In poker, we are almost always very concerned with the contents of our opponents' hands. But it is seldom possible to narrow down our estimate of these contents to a single pair of cards. Instead, we use a probability distribution to represent the hands he could possibly hold and the corresponding probabilities that he holds them. At the beginning of the hand, before anyone has looked at their cards, each player's probability distribution of hands is identical. As the hand progresses, however, we can incorporate new information we gain through the play of the hand, the cards in our own hand, the cards on the board, and so on, to continually refine the probability estimates we have for each possible hand.

Sometimes we can associate a numerical value with each element of a probability distribution. For example suppose that a friend offers to flip a fair coin with you. The winner will collect $10 from the loser. Now the results of the coin flip follow the probability distribution we identified earlier:

C = {(heads, $^1/_2$), (tails, $^1/_2$)}

Since we know the coin is fair, it doesn't matter who calls the coin or what they call, so we can identify a second probability distribution that is the result of the bet:

C' = {(win, $^1/_2$), (lose, $^1/_2$)}

We can then go further, and associate a numerical value with each result. If we win the flip, our friend pays us $10. If we lose the flip, then we pay him $10. So we have the following:

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

$B = \{(+\$10, {}^{1}/_{2}), (-\$10, {}^{1}/_{2})\}$

When a probability distribution has numerical values associated with each of the possible outcomes, we can find the ***expected value (EV)*** of that distribution, which is the value of each outcome multiplied by its probability, all summed together. Throughout the text, we will use the notation **<*X*>** to denote "the expected value of X." For this example, we have:

$<B> = \left({}^{1}/_{2}\right)(+\$10) + \left({}^{1}/_{2}\right)(-\$10)$
$<B> = \$5 + (-\$5)$
$<B> = 0$

Hopefully this is intuitively obvious - if you flip a fair coin for some amount, half the time vou win and half the time you lose. The amounts are the same, so you break even on average. Also, the EV of declining your friend's offer by not flipping at all is also zero, because no money changes hands.

For a probability distribution *P*, where each of the *n* outcomes has a value x_i and a probability then p_i *then P's* expected value <*P*> is:

$$< P >= \sum_{i=1}^{n} p_i x_i \qquad \textbf{(1.11)}$$

At the core of winning at poker or at any type of gambling is the idea of maximizing expected value. In this example, your friend has offered you a fair bet. On average, you are no better or worse off by flipping with him than you are by declining to flip.

Now suppose your friend offers you a different, better deal. He'll flip with you again, but when you win, he'll pay you $11, while if he wins, you'll only pay him $10. Again, the EV of not flipping is 0, but the EV of flipping is not zero any more. You'll win $11 when you win but lose $10 when you lose. Your expected value of this new bet B_n is:

$<B_n> = \left({}^{1}/_{2}\right)(+\$11) + \left({}^{1}/_{2}\right)(-\$11)$
$<B_n> = \$0.50$

On average here, then, you will win fifty cents per flip. Of course, this is not a guaranteed win; in fact, it's impossible for you to win 50 cents on any particular flip. It's only in the aggregate that this expected value number exists. However, by doing this, you will average fifty cents better than declining.

As another example, let's say your same friend offers you the following deal. You'll roll a pair of dice once, and if the dice come up double sixes, he'll pay you $30, while if they come up any other number, you'll pay him $1. Again, we can calculate the EV of this proposition.

$<B_d> = (+\$30)\left({}^{1}/_{36}\right) + (-\$1)\left({}^{35}/_{36}\right)$
$<B_d> = \$\,{}^{30}/_{36} - \$\,{}^{35}/_{36}$

$<B_d> = -\$\,{}^{5}/_{36}$ or about 14 cents.

The value of this bet to you is about negative 14 cents. The EV of not playing is zero, so this is a bad bet and you shouldn't take it. Tell your friend to go back to offering you 11-10 on coin flips. Notice that this exact bet is offered on craps layouts around the world.

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

A very important property of expected value is that it is ***additive.*** That is, the EV of six different bets in a row is the sum of the individual EVs of each bet individually. Most gambling games - most things in life, in fact, are just like this. We are continually offered little coin flips or dice rolls - some with positive expected value, others with negative expected value. Sometimes the event in question isn't a die roll or a coin flip, but an insurance policy or a bond fund. The free drinks and neon lights of Las Vegas are financed by the summation of millions of little coin flips, on each of which the house has a tiny edge. A skillful poker player takes advantage of this additive property of expected value by constantly taking advantage of favorable EV situations.

In using probability distributions to discuss poker, we often omit specific probabilities for each hand. When we do this, it means that the relative probabilities of those hands are unchanged from their probabilities at the beginning of the hand. Supposing that we have observed a very tight player raise and we know from our experience that he raises if and only if he holds aces, kings, queens, or ace-king, we might represent his distribution of hands as:

H = {AA, KK, QQ, AKs, AKo}

The omission of probabilities here simply implies that the relative probabilities of these hands are as they were when the cards were dealt. We can also use the <X> notation for situations where we have more than one distribution under examination. Suppose we are discussing a poker situation where two players A and B have hands taken from the following distributions:

A = {AA, KK, QQ, JJ, AKo, AKs}
B = {AA,KK,QQ}

We have the following, then:

< A, B > : the expectation for playing the distribution A against the distribution B.
< A, AA|B > : the expectation for playing the distribution A against the hand AA taken from the distribution B.
<AA| A, AA|B > : the expectation for playing AA from A against AA from B.
< A, B > = *p*(AA) < A, AA|B > + *p(*KK***)*** < A, KK|B > + *p*(QQ) < A,QO| B> ...and so on.

Additionally, we can perform some basic arithmetic operations on the elements of a distribution. For example, if we multiply all the values of the outcomes of a distribution by a real constant, the expected value of the resulting distribution is equal to the expected value of the original distribution multiplied by the constant. Likewise, if we add a constant to each of the values of the outcomes of a distribution, the expected value of the resulting distribution is equal to the expected value of the original distribution plus the constant.

We should also take a moment to describe a common method of expressing probabilities, ***odds.*** Odds are defined as the ratio of the probability of the event not happening to the probability of the event happening. These odds may be scaled to any convenient base and are commonly expressed as "7 to 5," "3 to 2," etc. ***Shorter*** odds are those where the event is more likely: ***longer*** odds are those where the event is less likely. Often, relative hand values might be expressed this way: "That hand is a 7 to 3 favorite over the other one," meaning that it has a 70% of winning, and so on.

Odds are usually more awkward to use than probabilities in mathematical calculations because they cannot be easily multiplied by outcomes to yield expectation. True "gamblers" often use odds, because odds correspond to the ways in which they are paid out on their bets. Probability is more of a mathematical concept. Gamblers who utilize mathematics may use either, but often

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

prefer probabilities because of the ease of converting probabilities to expected value.

Key Concepts

- The probability of an outcome of an event is the ratio of that outcome's occurrence over an arbitrarily large number of trials of that event.
- A probability distribution is a pairing of a list of complete and mutually exclusive outcomes of an event with their corresponding probabilities.
- The expected value of a valued probability distribution is the sum of the probabilities of the outcomes times their probabilities.
- Expected value is additive.
- If each outcome of a probability distribution is mapped to numerical values, the expected value of the distribution is the summation of the products of probabilities and outcomes.
- A mathematical approach to poker is concerned primarily with the maximization of expected value.

Read more poker books for free at: www.PokerBooks.LT